

Civics Group	Index Number	Name (use BLOCK LETTERS)
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ST. ANDREW'S JUNIOR COLLEGE
2008 JC2 Prelims

H2 BIOLOGY

9747/3

Paper 3: Applications

Monday

8 September 2008

1 hour 30 minutes

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/15
2	/15
3	/15
4	/20
Total	/65

This document consists of **10** printed pages.

[Turn over

Answer **all** the questions in spaces provided.

Question 1

Babies born with severe combined immune deficiency (SCID) have no defence against common infections and quickly become ill when the protection from maternal antibodies is lost. This was the result of a mutation in their ADA gene that resulted in infants suffering a deficiency in all functions of their immune system.

- (a) Explain briefly how a mutation in the ADA gene leads to “deficiency” in the functions of the immune system.

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Gene therapy for SCID has been carried out using the procedure shown in Fig. 1.1.

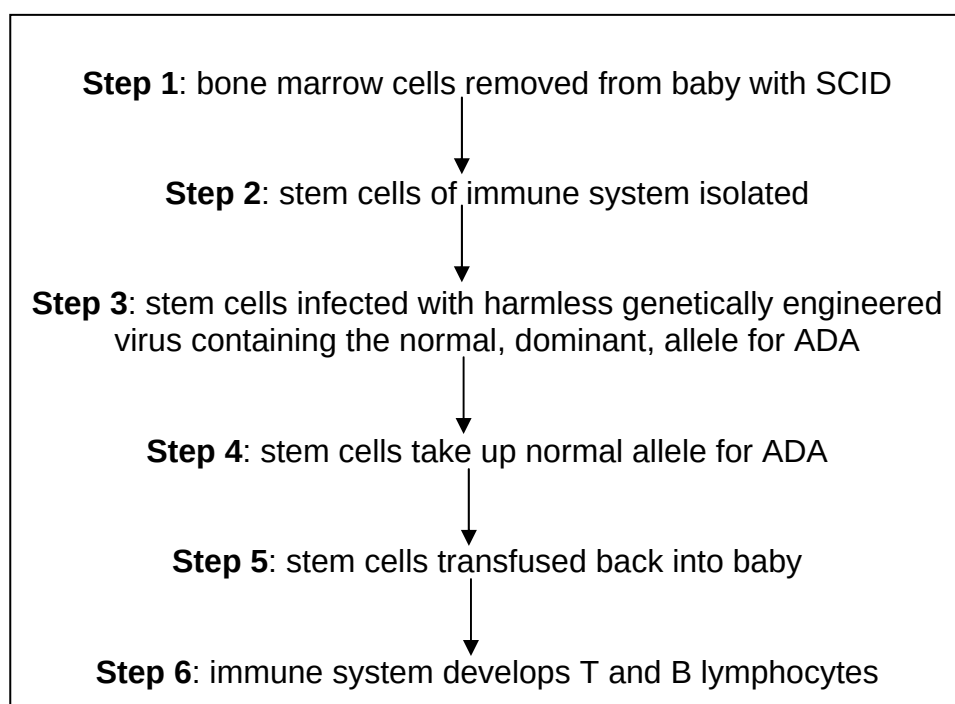


Fig. 1.1

- (b) (i) Describe the properties of the stem cells used in Fig. 1.1.

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(ii) List an advantages and disadvantage of using the human stem cells mentioned in Fig. 1.1.

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(c) (i) Explain what is meant by *gene therapy*.

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(ii) Why is it easier to perform gene therapy when the normal allele is the dominant allele of the gene concerned (step 3)?

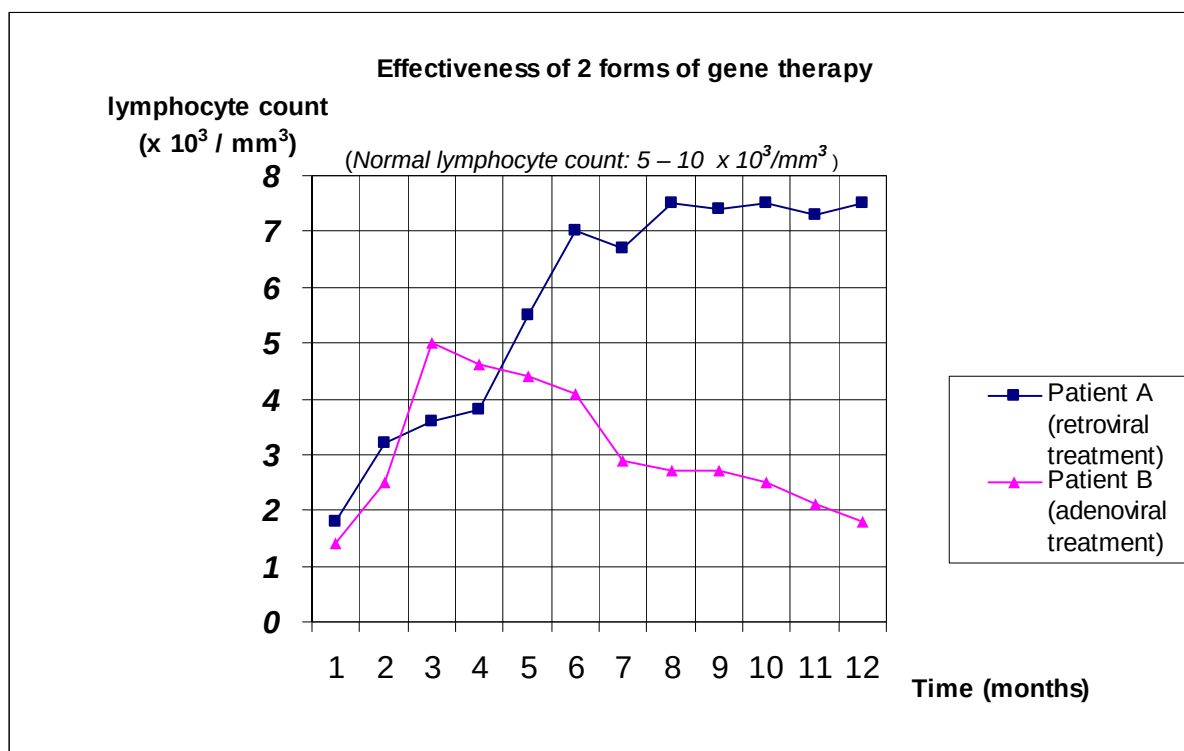
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The diagram below shows the lymphocyte counts (both T & B) of 2 infant patients that underwent 2 separate types of gene therapy treatments for SCID. Patient A was treated with a retrovirus while Patient B was treated with an adenovirus. The healthy lymphocyte count of the patients was monitored by a doctor at the end of every month for a period of 12 months. (The acceptable range of lymphocyte count for non-SCID patients is $5-10 \times 10^3/\text{mm}^3$).



With reference to diagram,

(d) (i) Describe the trends observed in the 2 patients' lymphocyte count.

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(ii) Account for the difference in lymphocyte count at the end of 12 months.

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[Qn 1 – Total: /15]

Question 2

A 5% herbicide solution of cyanamide kills the weeds which infest fields of tobacco plants, but causes some damages to the tobacco. However, tobacco plants have been genetically engineered by inserting a gene from a soil fungus, *Myrothecium verrucaria*. The gene codes for an enzyme, cyanamide hydratase, which converts cyanamide into urea.

The gene was inserted into a plant virus and then tobacco plants were infected with the virus. A strain of transgenic tobacco plants was thus developed that carried the gene for cyanamide hydratase but were unaffected by other viral proteins. The enzyme was most active in the leaves, showing about one fifth of the activity found in *Myrothecium*.

The transgenic plants thrive when treated with a dilute solution of cyanamide, since the plants have urease enzymes which convert the urea to ammonium ions which provide the plants with a source of nitrogen.

(a) Explain why the tobacco plants described in the passage are *transgenic plants*.

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(b) Why was the gene for cyanamide hydratase inserted into a virus?

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(c) Describe briefly another way of transforming plant cells other than the method used in this experiment.

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Equal masses of tobacco plant callus tissue were cultured for four weeks on media containing different concentrations of two plant growth regulators: auxin and cytokinin. The results are shown in Table 2.1 below.

treatment	concentration of plant growth regulators / mg dm ⁻³		effect of plant growth regulators on callus growth
	auxin	cytokinin	
A	2.00	0.00	little or no growth
B	2.00	0.02	growth of roots
C	2.00	0.20	increased growth of callus with no differentiation
D	2.00	0.50	growth of shoots
E	0.00	0.20	little or no growth

Table. 2.1

(d) Explain what is meant by *callus tissue*.

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(e) With reference to Table 2.1,

(i) Describe the effects of auxin and cytokinin on callus growth.

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[4]

(ii) Suggest what treatment should be given to the shoots developed from callus in treatment D to turn them into plantlets.

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[2]

Subsequently, extensive field trials were carried out to assess the environmental effects of growing these transgenic tobacco plants.

(f) (i) Give 3 reasons why it is important to assess the environmental effects of growing these transgenic tobacco plants.

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(ii) Other than providing resistance to herbicide, explain another way in which genetic engineering of tobacco plants could increase crop yield.

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[Qn 2 – Total: /15]

Question 3

RFLP analysis has facilitated the process of DNA fingerprinting and disease detection played an important role in the Human Genome Project.

One of its application, DNA fingerprinting, has been widely used to distinguish between individuals using only samples of their DNA. Forensic scientists can make DNA fingerprints from materials such as blood, skin and semen to identify criminals.

(a) Outline how genetic fingerprinting is carried out.

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(b) Explain why VNTRs are useful genetic makers for DNA fingerprinting whereas genes are seldom used for DNA fingerprinting.

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- (c) There are still genetic diseases for which no gene has yet been discovered. Until the gene can be located, cloned, and sequenced, no probe can be made to detect it directly. However, alleles for Huntington's disease and a number of other genetic diseases could be detected in an indirect way using RFLP. Explain how this can be done.

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- (d) How did RFLPs play an important role in the Human Genome Project?

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[Qn 3 – Total: /15]

Write your answer to this question on the separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 **(a)** Distinguish between genomic DNA and cDNA library. [6]
- (b)** Outline the procedures for the cloning of a named eukaryotic gene using a plasmid. [8]
- (c)** Discuss the difficult ethical concerns of the human genome project. [6]

[Total: 20]